

Graph pebbling is based on a game in which pebbles are moved across the vertices of a graph according to certain rules. Traditionally, the goal is to ensure that at least one pebble reaches a chosen target vertex. A more restrictive variation was recently introduced in 2024, where the objective is to place exactly one pebble on the target. [1] This idea leads to the concept of the *singular pebbling number*.

In both the classical and singular pebbling settings, the game is played between two roles: the **Placer**, who chooses how the pebbles are initially distributed and selects the target vertex, and the **Pebbler**, who must use the rules of the game to move pebbles toward the target and win. All graphs in this paper are assumed to be connected and simple (no loops or multiple edges between pairs of vertices).

Definition 1. A **configuration** of pebbles on a graph G is an assignment of a non-negative integer number of pebbles to each vertex.

Definition 2. A **pebbling move** removes two pebbles from a vertex v with at least 2 pebbles and places one pebble on an adjacent vertex u . The second pebble is then discarded.

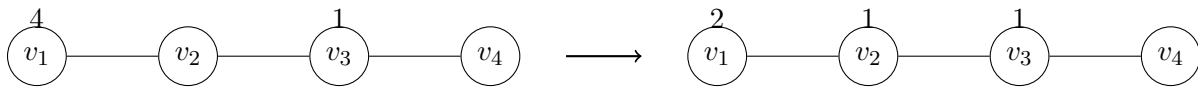


Figure 1: Illustration of a pebbling move

Definition 3. The **pebbling number** $\pi(G)$ of a graph G is the smallest number t such that, from any configuration of t pebbles, it is possible to move at least one pebble to any target vertex.

Definition 4. The **singular pebbling number** $\pi_s(G)$ is the smallest number t such that, from any configuration of at least t pebbles, it is possible to move exactly one pebble to the target vertex.

Note: The use of “at least t ” in the definition of the singular pebbling number is intentional. In contrast to traditional pebbling, having more pebbles does not hurt the pebbler’s chances of winning. But in singular pebbling, it can make it harder to succeed.

Definition 5. A graph is **complete** if there is an edge between every pair of distinct vertices. The complete graph on n vertices is denoted K_n .

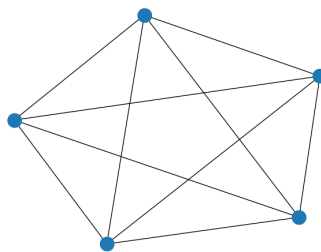


Figure 2: The complete graph K_5

Lemma 1. [1] *For the Placer to win, they cannot put three or more pebbles on the target vertex unless the graph is K_1 .*

Proof. If there is an odd number of pebbles on the target vertex, all but one of the pebbles can be moved to an adjacent vertex, leaving exactly one pebble on the target vertex.

If there is an even number of pebbles on the target vertex, all the pebbles can be moved to an adjacent vertex. There are therefore two pebbles on a vertex adjacent to the target vertex that can be moved to the target vertex, leaving exactly one pebble on the target vertex. $\square$

Definition 6. An *optimal configuration* is a placement of pebbles by the Placer that prevents the Pebbler from succeeding in placing exactly one pebble on the target vertex.

Question 1. *How do optimal configurations behave for larger or more intricate graphs?*

Example 1. [1] $\pi(K_2) = 2$, $\pi_s(K_2) = 3$.

Proof. First, let us determine the pebbling number. In order for the Placer to win, pebbles can only be placed on the vertex adjacent to the target vertex. If the Placer places more than one pebble on the adjacent vertex, the Pebbler will win in one movement toward the target vertex. Therefore, the pebbling number of K_2 is two.

Next, we determine the singular pebbling number. In order for the Placer to win, the options are to either place one pebble on the vertex adjacent to the target vertex or place two pebbles on the target vertex. Therefore, the singular pebbling number of K_2 is three. $\square$

Question 2. [2] *Is there an efficient way to compute $\pi(G)$ & $\pi_s(G)$ for arbitrary graphs, or is the problem computationally hard?*

Question 3. *What happens if we change the rules slightly — for instance, allow moving one pebble at a cost of two from a vertex to a non-adjacent one?*

References

- [1] Harmony R. Morris, *On the Singular Pebbling Number of a Graph*, Rose-Hulman Undergraduate Mathematics Journal, Volume 25, Issue 1 (2024). Available at: <https://scholar.rose-hulman.edu/rhumj/vol25/iss1/1/>
- [2] M. Yang, C. Yerger, and R. Zhou, *Lollipop and cubic weight functions for graph pebbling*, Journal of Combinatorial Optimization, vol. 49, no. 16, 2025. Available at: <https://doi.org/10.1007/s10878-024-01248-1>